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Resolving conflicts between data source updates and user edits to ontology data

Abstract

A system for resolving conflicts between data source updates and user edits to an ontology before applying the resulting changes to the ontology and related methods are disclosed. The system is programmed to receive data source updates from data sources and transform the data source updates to updates to ontology data. The system is also programmed to receive edits to ontology data from user accounts. The system is programmed to review these updates or edits to the ontology and resolve conflicts according to a predetermined strategy, such as prioritizing a user edit over a data source update. The resulting changes are incorporated to one or more databases where representations of the ontology data are stored.

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Background/Summary

BENEFIT CLAIM (1) This application claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Application No. 63/461,143, titled “ONTOLOGY ARCHITECTURE” and filed on Apr. 21, 2023, the entire contents of which are hereby incorporated by reference as if fully set forth herein.

TECHNICAL FIELD

(1) The present disclosure relates to ontology-based database operations, and more particularly to database operations related to indexing, conflict resolution, versioning, and access control.

BACKGROUND

(2) Data originating from data sources can be organized or structured according to an ontology, which facilitates understanding of and access to the data. The ontology can be backed by databases where the data is stored. It would be helpful to properly manage the databases while fully supporting operations of the ontology.

SUMMARY

(3) The appended claims may serve as a summary of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Example embodiments will now be described by way of non-limiting examples with reference to the accompanying drawings, in which:

(2) FIG. 1 illustrates an example networked computer system in which various embodiments may be practiced;

(3) FIG. 2 illustrates an example set of components of and workflows managed by a computer application server (“server”).

(4) FIG. 3 illustrates an example view of an indexing pipeline implemented by the server via the object data funnel.

(5) FIG. 4 illustrates an example framework for resolving conflicts between data source updates and user edits using the conflict resolution strategy that prioritizes user edits.

(6) FIG. 5A illustrates part of an example process of resolving conflicts between data source updates and user edits using the conflict resolution strategy that prioritizes user edits.

(7) FIG. 5B illustrates part of an example process of resolving conflicts between data source updates and user edits using the conflict resolution strategy that prioritizes user edits.

(8) FIG. 5C illustrates part of an example process of resolving conflicts between data source updates and user edits using the conflict resolution strategy that prioritizes user edits.

(9) FIG. 6 illustrates an example graphical user interface for creating a granular policy.

(10) FIG. 7 illustrates an example process of indexing data changes to ontology-based databases in accordance with disclosed embodiments.

(11) FIG. 8 illustrates a process of resolving conflicts in changes to ontology data in accordance with disclosed embodiments.

(12) FIG. 9 illustrates a process of managing access control for ontology data in accordance with disclosed embodiments.

(13) FIG. 10 illustrates a computer system upon which various embodiments may be implemented.

DETAILED DESCRIPTION OF CERTAIN EMBODIMENTS

(14) In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the example embodiment(s) of the present invention. It will be apparent, however, that the example embodiment(s) may be practiced without these specific details. In other instances, well-known structures and devices are shown in block diagram form in order to avoid unnecessarily obscuring the example embodiment(s).

1. General Overview

(15) A system for resolving conflicts between data source updates and user edits to an ontology before applying the resulting changes to the ontology and related methods are disclosed. The system is programmed to receive data source updates from data sources and transform the data source updates to updates to ontology data. The system is also programmed to receive edits to ontology data from user accounts. The system is programmed to review these updates or edits to the ontology and resolve conflicts according to a predetermined strategy, such as prioritizing a user edit over a data source update. The resulting changes are incorporated to one or more databases where representations of the ontology data are stored.

(16) In some embodiments, the system is programmed to manage the ontology, which includes definitions of ontology entity types, and ontology entities that are instantiated from the ontology entity types. An ontology entity type is generally an object type or a link type and has one or more properties. For example, an ontology object type can correspond to a person and has properties corresponding to a person's residence and occupation, and an ontology link type can correspond to a relationship between two people.

(17) In some embodiments, the system is programmed to receive original data from the data sources. Each data source can be represented as a dataset that has a specific schema, which can be mapped to specific ontology entity types to enable transformation of the original data from the data source to ontology entities of the specific ontology entity types. For example, a dataset can have columns that correspond to a person's home address and job, and those columns could be mapped to certain properties of the person object type.

(18) In some embodiments, in addition to updates from the data sources, the system is programmed to also receive edits to the ontology from user accounts, including requests to create or remove ontology entities or view or modify ontology entities. The system is programmed to review the updates or edits to the same ontology entity in a chronological order and resolve conflicts between the two sources of change according to a predetermined strategy. The system is programmed to apply the same strategy to the same ontology entity type over time or across ontology entities but could apply different strategies to different ontology entity types.

(19) One predetermined strategy prioritizes a user edit to a data source update. Such a strategy can indicate that when a property of an ontology entity is being changed to a first value via a data source update but to a second value via a user edit, then the property will be set to the second value. For example, at a certain time, when the job property of a first person object is being changed to "Nurse" via a data source update and to "Teacher" via a user edit, the job property of the first person object will be set to "Teacher". Such a strategy can further indicate that after an ontology entity is created via a user edit, subsequent changes to the ontology entity via data source updates will be ignored. For example, a second person object having a certain value for the primary key may have originate from a data source and later deleted via a data source update. When the second person object is subsequently recreated with the same value for the primary key via a user edit, future changes to the second person object via data source updates will be ignored. The conflict-free changes to the ontology can then be represented in the one or more databases. Another predetermined strategy prioritizes a later change to an earlier change.

(20) The system disclosed herein has several technical benefits. An ontology provides a structure to any large volume of data, which facilitates user interaction with the data. By updating the one or more databases with only changes to the ontology data, the system reduces the amount of data being incorporated into the one or more databases and thus improves the network traffic and

database performance. By resolving conflicts between the two sources of change to the ontology before updating the one or more databases, the system improves data quality and further reduces the amount of data being incorporated into the one or more databases. By supporting a strategy that prioritizes a user edit over a data source update, the system provides a user-centric view of the evolution of the ontology that is consistent with user actions.

2. Example Computing Environments

(21) FIG. 1 illustrates an example networked computer system in which various embodiments may be practiced. FIG. 1 is shown in simplified, schematic format for purposes of illustrating a clear example and other embodiments may include more, fewer, or different elements.

(22) In some embodiments, a networked computer system **100** comprises a computer server (“server”) **102**, one or more user devices **130**, and one or more data sources **140**, which are communicatively coupled through direct physical connections or via a network **118**.

(23) In some embodiments, the server **102** is programmed or configured to manage datasets representing data sources, which are transformed to ontology data, and databases representing the ontology data in multiple forms. The management comprises integrating multiple sources of change to the ontology data, representing changes to the ontology data in different forms, and allow access to the ontology data in various granularities. The server **102** can comprise any centralized or distributed computing facility with sufficient computing power in data processing, data storage, and network communication for performing the above-mentioned functions.

(24) In some embodiments, each user device of the one or more user devices **130** is programmed to submit access requests to the server **102** to access the ontology and receive replies to the access requests from the server **102**. The access requests can include reading, writing, adding, or deleting ontology data. Each user device can comprise a personal computing device, such as a desktop computer, laptop computer, tablet computer, smartphone, or wearable device.

(25) In some embodiments, each data source of the one or more data sources **140** is programmed to provide original data to the server **102**, which represents the original data as datasets, or accept changes to the original data based on updates to the datasets from the server **102**. Each data source can generally be similar to the server **102** and comprise any computing facility with sufficient computing power in data processing, data storage, and network communication for performing the above-mentioned functions.

(26) The network **118** may be implemented by any medium or mechanism that provides for the exchange of data between the various elements of FIG. 1. Examples of the network **118** include, without limitation, one or more of a cellular network, communicatively coupled with a data connection to the computing devices over a cellular antenna, a near-field communication (NFC) network, a Local Area Network (LAN), a Wide Area Network (WAN), or the Internet, a terrestrial or satellite link.

(27) In some embodiments, the server **102** is programmed or configured to receive original data from the one or more data sources **140**. The server **102** is programmed to set up datasets representing the one or more data sources **140** and transforming data in the datasets to ontology data. The server **102** is programmed to further represent the ontology data in multiple databases. On the other hand, the server **102** is programmed to receive access requests from the one or more user devices **130**, including user edits to the ontology data. The server **102** is programmed to also integrate these user edits into the multiple databases. The server **102** can also be programmed to materialize these user edits into datasets and transform data in these datasets for integrating into the datasets representing the one or more data sources **140** and ultimately updating the one or more data sources **140**.

3. Functional Descriptions

(28) FIG. 2 illustrates an example set of components of and workflows managed by the server **102**. In some embodiments, the server **102** is programmed to receive original data from various data sources and maintain them as datasets **202**. The data sources can include file systems, database

management systems, streaming sources, or blob stores, for example. A dataset is a wrapper around a collection of files which are stored in a specific file system. The datasets **202** can contain mainly structured and semi-structured data, from which one or more schemas can be inferred. Each schema specifies how a file is to be parsed into a specific format, such as a tabular format having rows and columns.

(29) In some embodiments, the server **102** is programed to establish a mapping between the group of datasets and an ontology via the ontology management **204** as a service. The ontology includes definitions for all relevant object types, such as people, computers, networks, documents, or events, with objects being instantiated from the object types. The ontology can also separately include definitions for all relevant link types, such as similarities or dependencies, with links being instantiated from the link types. Specific structures, such as properties or versions, and specific operations, such as revision, access control, or provenance tracking at the object, link, or property level (or corresponding row or column level) apply to all object types and all link types. The object types and link types are considered as “ontology entity types”. The objects and the links are considered as “ontology entities”, which encompass the properties. The ontology and relevant metadata are considered as “ontology data”. The ontology management **204** can manage definitions for an application programming interface (API) that allows access to the ontology entities and the structures. The mapping between the group of datasets and the ontology would associate, for each dataset, one or more ontology entity types to which the dataset can be transformed. The server **102** can be further programmed to establish, for each dataset, a transformation between a schema for the dataset and definitions for the one or more ontology entity types to which the dataset is associated. For example, a column in the schema can be transformed to a property of an object type and vice versa. When more original data is received from the various data sources and maintained as updated datasets, any updates in the updated datasets can be readily converted to ontology data according to the transformation.

(30) In some embodiments, the server **102** is programmed to store representations of ontology entity types and thus ontology entities in one or more ontology databases **206** (also called “object databases”) via an ontology data funnel **208** (also called “object data funnel”) as a service. The object data funnel **208** implements an indexing process of making the datasets as transformed to corresponding ontology data available in the object databases **206** for fast retrieval in response to user queries. The fast retrieval is achieved by the organization of the ontology, the representation of the ontology data in multiple object databases having different capabilities, and the availability of indices of ontology data and a mapping between the ontology and the object databases **206**.

(31) As links connect objects, link information could be considered as part of an object and vice versa. Therefore, representation of link types and that of object types can affect each other. In some embodiments, the server **102** is programmed to represent an object type or link type in different ways in different object databases in accordance with the different capabilities of the different object databases. As one example, in a relational object database, an object may be represented as a row while a property of an object may be represented as a column, which enables fast identification of objects that have certain property values. As another example, in a graph object database, an object may be represented as a node and a link between two objects may be represented as an edge, which enables fast exploration of relationships or links among objects. In other examples, the object databases can specialize in representing a property in a compressed form or an encrypted form, or representing properties in a specific order to form an index.

(32) In some embodiments, the server **102** can be programed to establish the mapping between the object databases and the ontology also via the ontology management **204**. This mapping would associate, for each ontology entity type, one or more object databases to which the ontology entity type can be transformed. The server **102** can be further programmed to establish, for each ontology entity type, a transformation between a definition for the ontology entity type and one or more schemas for the one or more object databases to which the ontology entity type is associated. For

example, a property of an object type can be transformed to a column in a schema and vice versa. Subsequently, any changes to the ontology can be readily converted to changes in the object databases **206** according to this transformation.

(33) In some embodiments, the server **102** is programmed to represent each ontology entity type in at least one of the object databases **206**, with the one being considered as the canonical object database for the ontology entity type. The server **102** can be programmed to also represent the ontology entity type in one or more secondary object databases. As changes are made to the ontology data, the indices could be updated as part of the indexing process implemented by the object data funnel **208**. In addition, the server **102** can be programmed to represent each ontology entity in different versions in the object databases **206** and preserve versioning consistency across the object databases **206**. The availability of a particular version in the canonical object database can be used to determine whether a particular version of an ontology entity is to be served in response to user queries.

(34) In some embodiments, the server **102** is programmed to manage not only objects or links including components originating from only one data source but also objects including components originating from multiple data sources. Each ontology entity that originates from multiple data sources is referred to as a multi-data source ontology entity (“MDO”). An MDO typically has these components that correspond to respective data sources and are separately represented in the object databases **206**. For example, separate components could be stored as separate columns in an object database. The server **102** can be further programmed to control access to the ontology data, from the ontology level down to the component level. Such access control can be extended to the object databases **206** where the ontology data are represented.

(35) In some embodiments, the server **102** is programmed to receive queries from a user device associated with a user account and execute the query via the ontology set **220** (also called “object set”) as a service. The object set **220** can support the API for querying the ontology noted above. The API would include a set of functions that operate on ontology entity types or ontology properties, such as filters or aggregators. These functions generally correspond to read requests to view ontology data, which require read access to the object databases **206**. These functions can also be called to process ontology data before actions are applied, as discussed below. The actions correspond to write requests to modify ontology data, which require write access to the object databases **206**.

(36) In some embodiments, the server **102** is programmed to receive an action comprising user edits **210** to ontology data from a user device, and apply the action via the action application **212** as a service. In this ontology context, an action specifies user logic to change data related to one or more ontology entities and is to be performed as a single transaction. In other embodiments, the server **102** is programmed to receive updates to the datasets **202** from the data sources. As noted above, when such updates are received, they can be readily converted to corresponding changes to the ontology data according to the transformation.

(37) In some embodiments, the server **102** is programmed to store representations of ontology data in the object databases **206** via the object data funnel **208**, as discussed above. The object data funnel **208** can implement the indexing process by indexing only changes and sending the indexed changes to the object databases **206**. As the changes can originate from the user edits **210** or data source updates (that turn into dataset updates), the server **102** can be programmed to resolve any conflicts in the received changes before indexing the changes and sending the indexed changes via the object data funnel **208** to the object databases **206**.

(38) 3.1. Processing User Edits to Ontology Data

(39) In some embodiments, the server **102** is programmed to allow users to make changes to ontology data as user edits by creating actions, as discussed above. An action specifies user logic to make these changes and is to be performed as a single transaction. For example, an action type that defines an Assign Employee action can specify changing the Role property value for a given

Employee object based on user input and creating a link between the Employee object and a new Manager object based on specific rules. As illustrated in this example, an action can include a series of user edits, with each corresponding to one change, such as updating the value of one property or deleting an object. Each user edit can be represented as a record that indicates a primary key identifying an object or a link in the ontology, an order, change information describing what is being changed, or other relevant data.

(40) In some embodiments, the server **102** is programmed to receive a request to apply an action to change ontology data from a user device. The server **102** can be programmed to store the request in an action queue together with a request timestamp, such as coordinated universal time (UTC) offset, to support user edits included in multiple actions to be applied. The request timestamp is associated with all the user edits included in the action. The server **102** can also be programmed to receive requests to read or view ontology data and store these requests together with request timestamps. All requests are processed based on the corresponding request timestamps. This ensures that, for example, when an object read occurring as part of an ontology query is requested after an object modification is made, the result of the object read is guaranteed to reflect the user edits in the user modification. The server **102** is further programmed to dequeue the requests at least from the action queue continuously or based on a predetermined schedule or triggers.

(41) 3.2. Indexing Data Changes to Object Databases

(42) FIG. 3 illustrates an example view of an indexing pipeline implemented by the server **102** via the object data funnel **208**. FIG. 3 is shown in simplified, schematic format for purposes of illustrating a clear example and other embodiments may include more, fewer, or different elements connected in various manners, or different values within the different elements.

(43) 3.2.1. Creating Changelog Datasets for Data Source Updates

(44) In some embodiments, the server **102** is programmed to receive new original data from a data source and compute the difference or change from what is already in the data source via the corresponding dataset. For example, when the dataset contains rows and columns, the difference could be an addition, removal, or update of a row (with a new value for a column). The server **102** is programmed to then create a changelog dataset. The changelog dataset can indicate, for each difference, a primary key of the affected row, an order (e.g., a change timestamp), change information describing what is being changed, or other relevant data. For example, in FIG. 3, a changelog dataset contains a row **302** having “X2” as the value for the Seq column, and “Nurse” as the updated value for the Work column. The server **102** can be programmed to maintain a single changelog dataset at a time or one changelog dataset for each data source. Each changelog dataset can be developed through append transactions that add new files indicating the computed differences.

(45) 3.2.2. Merging Data Source Updates and User Edits

(46) In some embodiments, the server **102** is programmed to combine data source updates and user edits, which constitute two sources of change to the object databases **206**. The data source updates can come from a changelog dataset, and the user edits can come from an action queue. For example, in FIG. 3, a set of user edit records in the action queue contains a record **304** having “Y3” as the value for the ID field and “JS” as the updated value for the Name field. This record corresponds to a request to update the value of the Name property of a Person object with a primary key of “Y3” for the ID property in the ontology. The user edit records can also be stored into a user edit dataset for downstream processing. The server **102** can be configured to perform this combination or merge of data source updates and user edits based on a predetermined schedule or triggers. Example triggers include when a certain amount of data in either source of change has been received or detected or when a certain period of time has passed. It is possible that when a trigger runs, only one of the sources of change is available for the merge.

(47) In some embodiments, the server **102** is programmed to merge the data source updates and the user edits into a merged dataset that enables efficient indexing into the object databases **206**. While

the user edits would generally refer to ontology data, such as an object ID, the data source updates involve original data in the data sources. The server **102** thus can be programmed to convert the changelog dataset into a transformed dataset that refers to the ontology data using the transformation discussed above. For example, in FIG. 3, the transformed dataset corresponding to the changelog dataset contains a row **306** having “Y3” as the value for the ID column and “Nurse” as the updated value for the Job column, which respectively correspond to the ID property and the Job property of a Person object.

(48) In some embodiments, the server **102** is programmed to join the transformed dataset and the set of user edit records (or the user edit dataset) into a merged dataset based on primary keys each identifying an object or a link in the ontology. For example, in FIG. 3, after the transformation, since the row **306** and the record **304** have a common value for the primary key, they can be joined together to create the merged dataset that has a row **308** having “Y3” as the value for the ID column, “JS” as the updated value for the Name column, and “Nurse” as the updated value for the Job column. The merged dataset can be incorporated back into the data sources. The merged dataset can contain one row for each ontology entity that includes all the compatible changes being made to the ontology entity. When certain changes respectively from the two sources of change are conflicting and thus incompatible, the server **102** is configured to resolve the conflict, as further discussed in Section 3.3.

(49) 3.2.3. Creating Index Data

(50) In some embodiments, the server **102** is programmed to create an index dataset from the merged dataset for at least one of the object databases **206**. These index datasets enable fast retrieval of representations of ontology data in the object databases **206**. As discussed above, each ontology entity type can be represented in multiple object databases. The server **102** can be configured to first consult the mapping that indicate how each ontology entity type covered in the merged dataset is represented in multiple object databases. The server **102** can be further programmed to determine for which object databases the index datasets need to be generated from the merged dataset. As each object database has its set of capabilities and its approach of organizing data, the index datasets for different object databases can look different. For each object database, the server **102** can be configured to then consult the database schema and then create a corresponding index dataset in the format required by the object database.

(51) In some embodiments, an index dataset comprises indices of the changes captured in the merged dataset to ontology entities that are represented in the corresponding object database. Each index can include a list of identifiers of the objects or links undergoing changes and the corresponding changes. The index dataset can additionally include a pointer to a specific location in the appropriate dataset for each of the affected object or link. For example, an index in an indexed dataset can indicate that for the Job property, a first object that has an updated value of “Nurse” has “Y3” as the value for the ID property and a second object that has the same updated value has “W9” as the value for the ID property. This index thus communicates the changes and also enables an efficient search of the ontology based on the Job property of the person object type. The index could also point to the dataset that gives rise to the object, which would contain the entire row corresponding to the object. In FIG. 3, the merged dataset leads to the first indexed dataset that facilitates finding an object that is represented as a row **310** having “JS” as the value for the Name column and “Nurse” as the value for the Job column in a first object database that is a relational database, and a second index dataset that facilitates finding t—the same object that is represented as a node **312** with having “JS” as the value for the Name attribute and “Nurse” as the value for the Job attribute in a second object database that is a graph database.

(52) In some embodiments, the server **102** is programmed to apply an action immediately by dequeuing the request to apply, creating corresponding index data directly from the user edits in the action, and transmitting the index data to one or more object databases in real time. This process bypasses the persistence of the user edits via the user edit dataset or the merged dataset via the

object data funnel **208**. However, user edits are also persisted periodically, as discussed above. Therefore, index data can be created directly from the user edit records or from intermediary index datasets or other temporary files, based on the database schema for each object database. As actions are applied immediately and indexed into the object databases continuously, the user edits to be persisted into a merged dataset can generally already be indexed into the object databases. The merged dataset and a corresponding index dataset generally would not conflict with the index data or representation of the ontology data in the object database as long as the data source updates do not take precedence over the user edits but are merely verified against the user edits.

(53) 3.2.4. Preparing Changed Ontology Data for Querying

(54) In some embodiments, the server **102** is programmed to transmit data from the index datasets to the appropriate object databases that can be queried. This approach can be analogized to the approach discussed in U.S. Pat. No. 11,169,507, where the updates from the data sources and the user edits can correspond to the log stream, the merged datasets can correspond to the hot storage associated with the indexers or the data stored in that hot storage, the index datasets can correspond to the indexed portions, and the object databases can correspond to the hot storage associated with the search nodes or the data stored in that hot storage. According to the approach discussed in the patent, some indexed portions can be stored in cold storage associated with the indexers, and a catalog can include only metadata or pointers to locations of the indexed portions in the cold storage instead of the actual indexed portions. The catalog is then consulted in response to a search request, and an appropriated indexed portion is then retrieved from the cold storage based on the consultation and transmitted to the hot storage associated with the search nodes. Therefore, while in some embodiments, the indexed datasets include the indexed portions as indices and are transmitted to the object databases according to a predetermined schedule or specific triggers, in other embodiments, some of the indices can be pushed to certain cold storage and transmitted to the object databases according to a similar catalog only in response to search requests. Example triggers for transmitting the index datasets include when a certain amount of data in one or more index datasets has become available or when a certain period of time has passed.

(55) 3.3. Resolving Conflicts Between User Edits and Data Source Updates

(56) As discussed above, data source updates and user edits constitute two sources of change to the object databases. When certain changes are conflicting and thus incompatible, the server **102** is configured to resolve the conflict. In some embodiments, when a specific object or link with a primary key value is subject to conflicting changes from the data sources and user edits in a merged dataset, the server **102** is programmed to resolve the conflicts with a conflict resolution strategy. The merged dataset can include multiple changes associated with the primary key and different timestamps. The server **102** can be programmed to go through these changes in a chronological order in applying the conflict resolution strategy. The merged dataset can thus include one row for each ontology entity with all the changes to be made.

(57) In some embodiments, the server **102** is programmed to adopt a conflict resolution strategy that prioritizes user edits. The strategy can enforce the following resolution rules. (1) Once an ontology entity is modified based on a user edit, the user edit remains effective until overridden by another user edit when the ontology entity is not being deleted. (2) Rule (1) holds except that a data source update to delete a corresponding row will be effective when the corresponding ontology entity was not created based on a user edit. (3) Rule (1) holds despite the ontology entity being possibly deleted and recreated again based on data source updates. (4) Once an ontology entity is created or deleted based on a user edit, all future data source updates to the ontology entity are ignored. Pursuant to these rules, the server **102** is programmed to consider data source updates to an ontology entity originally created based on a data source update until the object is deleted based on a user edit. Once a user edit is applied to a property of an ontology entity, the server **102** is programmed to receive further changes to the property only based on user edits.

(58) FIG. 4 illustrates an example framework for resolving conflicts between data source updates

and user edits using the conflict resolution strategy that prioritizes user edits. In some embodiments, for each latest user edit, at step **402**, the server **102** is programmed to determine whether this latest user edit is to delete an object from the ontology. In response to a positive determination, at step **404** the server **102** is programmed to conclude that the object is not visible in the ontology, regardless of whether any corresponding row is in one of the data sources. In response to a negative determination, which means that the latest user edit is to create or modify an object in the ontology, at step **406**, the server **102** is programmed to determine whether there was a user edit to delete the object. In response to a positive determination, which means that the user edit is to recreate the, at step **408**, the server **102** is programmed to conclude that the object is visible in the ontology, and ignore future data source updates to the object. In response to a negative determination, at step **410**, the server **102** is programmed to determine whether there has been a user edit to create the object. In response to a positive determination, which means that the user edit is to initially create the object (after a data source update to delete the object), the server **102** similarly reaches step **408**. In response to a negative determination, which means that the latest user edit is to modify an object that was not last created based on a user edit, at step **412**, the server **102** is programmed to determine whether the object is present in the database. In response to a positive determination, which means that the object was created based on a data source update, at step **414**, the server **102** is programmed to conclude that the object is visible in the ontology, and ignore future data source updates to properties that are being modified in the latest user edit. In response to a negative determination, which means that a modification is being attempted in the latest user edit to an object that has not been created or recreated in the ontology, at step **416**, the server is programmed to conclude that the object does not exist and thus is not visible in the ontology.

(59) FIGS. 5A, 5B, and 5C illustrate an example process of resolving conflicts between data source updates and user edits using the conflict resolution strategy that prioritizes user edits. In FIG. 5A, at time T₀, there is a data source update to set a row with values for the col1 and col2 columns. Therefore, an object identified by the primary key of “pk1” is created in the ontology with values for the col1 and col2 properties. At time T₁, there is a data source update to remove all columns. Therefore, the object is deleted from the ontology. At time T₂, there is a data source update to set a row with values for the col1 and col2 columns. Therefore, the object is recreated in the ontology. At time T₃, there is a user edit to modify the value of the col2 property. Therefore, the value of the col2 property of the object is changed in the ontology. At T₄, there is a data source update to remove all columns. Therefore, the object is deleted from the ontology pursuant to Rule (2) noted above. At T₅, there is a data source update to set a row with values for the col1 and col2 columns as well as the user edit made at T₃ pursuant to Rules (1) and (3) noted above. Therefore, the conflict involving the property col2 is to be resolved by prioritizing user edits. The object is then recreated in the ontology with values for the col1 and col2 columns, where the value comes from the user edit. At time T₆, there is a data source update to modify the value of the column col1 of the row. Therefore, the value of the col1 property of the object is changed.

(60) In FIG. 5B, at time T₇, there is a user edit to delete the object, and the user edit made at time T₃ is no longer effective pursuant to Rule (1) noted above. Therefore, the object is deleted from the ontology. At time T₈, there is a data source update to modify the value of the col3 column of the row against the user edit made at T₇. Therefore, the conflict involving all the properties is to be resolved by prioritizing user edits. The object then remains deleted from the ontology pursuant to Rule (4) noted above. At time T₉, there is a user edit to create the object with a value for the col3 property. Therefore, the object is recreated with a value for the property col3. At time T₁₀, there is a data source change to modify the value of the col2 column of the row against the user edit made at time T₉. Therefore, the conflict involving the col3 property is to be resolved by prioritizing user edits. The object then remains unchanged in the ontology pursuant to Rule (4). At time T₁₁, there is a user edit to modify the value of the col2 property of the object. Therefore, the value of the col2 property is changed in the ontology. At time T₁₂, there is a data source update to delete all the

columns against the user edit made at T11. Therefore, the conflict involving all the properties is to be resolved by prioritizing user edits. The object then remains unchanged in the ontology pursuant to Rule (4).

(61) In FIG. 5C, at time T13, there is a data source update to create the row with values for the columns col1, col2, and col3 as well as a user edit to delete the object from the ontology. Therefore, the conflict involving all the properties is to be resolved by prioritizing user edits. Then object then is deleted from the ontology pursuant to Rule (4). At time T14, there is a user edit to modify the object without creating the object, which would fail. Therefore, the object remains deleted from the ontology.

(62) In some embodiments, the conflict resolution strategy is to prioritize data source updates, prioritize the more recent changes, or follow other predetermined logic. The server 102 can be programmed to adopt the same conflict resolution strategy across all object types. Alternatively, the server 102 can be programmed to adopt different conflict resolution strategies for different portions of the ontology, such as a specific conflict resolution strategy for a specific object type, a specific object type and all the sub-object types, or a specific object type and all the linked object types.

(63) 3.4. Controlling Ontology Data Permissions

(64) As discussed above, an ontology has objects and links, and each object or link has properties. In addition, each MDO has multiple components corresponding to multiple data sources, where each component includes properties that come from a single one of the multiple data sources. In certain embodiments, different components of an MDO include different properties, thus allowing no duplicity within the object.

(65) In some embodiments, the server 102 is programmed to control access to the ontology at different levels, such as the ontology level, link type level, link level, object type level, object level, component level, or property level. The access control can be defined with respect to a hierarchy of user roles and a set of permissions or access levels.

(66) In some embodiments, the server 102 is programmed to associate each data source with a set of user roles for users or user accounts and corresponding permissions. For example, an ontology owner role has full access including being able to manage permissions, an ontology editor role is less than the owner role by not being able to manage permissions, an ontology viewer role is less than the editor role by not having editing permissions, and an ontology discoverer role is less than the viewer role by only being to see select metadata. In general, each data source leads to a distinct set of object types and link types, and the access control of the data source can determine the access control of the distinct set of object types and link types. In the case of MDOs, multiple data sources lead to an ontology entity type, and the access controls of the data sources can determine the access controls of the multiple components of the ontology entity type. When MDOs are represented in the object databases, such as one with rows and columns, the access controls of the multiple components of each ontology entity type would correspond to column-level access controls.

(67) In some embodiments, the server 102 is programmed to require having permissions to view an entire ontology entity to be able to modify the ontology entity. Therefore, when a user account is granted access by a first data source but not a second data source, the user account would not have access to modify an MDO that originates from both the first data source and the second data source.

(68) In some embodiments, the server 102 is programmed to require permissions as follows for accessing objects of an object type, all properties of which originate from two data sources of D1 and D2. Creating an object of the object type by setting values for the properties that originate from D1 requires only permissions for viewing D1. The values for the properties that originate from D2 could be set to null. Similarly, modifying an object of the object type by setting values for the properties that originate from D1 requires only permissions for viewing D1. Deleting an object of the object type, however, requires permissions for viewing both D1 and D2.

(69) In some embodiments, the server is programmed to require permissions as follows for

accessing links of a link type that connects two object types. The first object type has all properties originating from D1, and the second object type has all properties originating from D2. Creating or deleting a link of the link type between a first object of the first object type and a second object of the second object type requires only permissions for viewing the first object and the second object. (70) In some embodiments, the server **102** is programmed to manage granular policies. With each data source leads to a distinct set of object types and link types, the access control of the data source can apply at not only these levels but also the object level, link level, object property level, or link property level with granular policies. A certain granular policy can restrict access to certain ontology entities to certain user accounts based on certain attribute values of the certain user accounts or certain properties of the certain ontology entities. As a result, only those user accounts having the certain attribute values and those ontology entities having the certain property values would meet the requirements of the certain granular policy, and only the certain user accounts would have access to the certain ontology entities or the certain properties of the certain ontology entities. For example, the certain granular policy can allow user accounts from Europe to see and edit European objects. When ontology entities are represented in the object databases, such as one with rows and columns, the access controls that apply at the ontology entity level would correspond to row-level access controls, and those that apply at the property level would correspond to column-level access controls.

(71) FIG. **6** illustrates an example graphical user interface for creating a granular policy. In some embodiments, a granular policy can comprise one or more rules combined by logical operators. An administrator, such as a user in the ontology owner role, can add a rule or logical operator using the buttons in the area **602**. For adding a rule, an initial mechanism of choosing two of three items for comparison can be employed in the area **604**, and some sample comparisons are listed in the area **606**. The initial mechanism allows three types of comparisons. The first type of comparison compares an attribute of a user account (“user attribute”) with a specific value, and the rule **606c** is an example. The second type of comparison compares a property of an ontology entity (“ontology entity property” corresponding to a column) with a specific value, and the rule **606d** is an example. The third type of comparison compares a user attribute with an ontology entity property, and the rule **606a** and the rule **606b** are examples. The administrator can choose two of the three items in the area **604**, specify a relational operator, such as “includes” or “is equal to”, and obtain an initial rule, such as the rule **606b**. The user can then instantiate each of the items, such as assigning “Group IDs” to the attribute item and “Group IDs” to the column item of the rule **606b**, to get the final rule, such as the rule **606a**, which indicates that the Group IDs user attribute includes a value that is not manually specified but comes from the “Group IDs ontology entity property.”

(72) In some embodiments, a final rule can then be displayed in the area **608**, and it can be combined with existing rules using one or more logical operators, such as “any”. The default access scope for a specific user in an appropriate user role when a rule is satisfied can be the objects that are compared in the rule or all objects otherwise. For example, the default access scope for the specific user from satisfying the rule **606a** can be all objects where the Group IDs property match the Group IDs attribute of the specific user. The default access scope for the specific user from satisfying the rule **606c** can be all objects. The default access scope for the specific user from satisfying the rule **606d** can be all objects where the specified ontology entity property is equal to the specified value. The default access scope for satisfying a combination of rules can be the intersection of the access scopes of the matched rules. Additional elements can be included in the area **602**, the area **604**, or another area in the graphical user interface for specifying the access scope when each rule or the combination of rules is satisfied, which can be limited to specific properties of the objects. For example, a certain access scope can be defined to exclude from access those object properties that are being compared in the rules and the values thereof for users in roles that have editing permissions, to reduce disruptions to the application of the rules.

(73) As discussed above, in some embodiments, the granular policies are applied on a per data

source basis. The access controls associated with data sources can take precedence over those based on granular policies. For example, an object type might have two properties A and B that come from different data sources. For a specific user, the access controls associated with the data sources can determine that the ontology entity property B is inaccessible. When relevant granular policies are applied, only the ontology entity property A is then available for comparison, and the application result can be that only the ontology entity property A of certain objects are accessible. The presentation of the application result can depend on the exact user roles and permissions. When an ontology entity property is accessible, it can mean that only the metadata of the ontology entity property (e.g., identifier) is accessible, or both the metadata and actual value are accessible. Therefore, in the example, the metadata of the ontology entity property A would be shown and the value of this ontology entity property would be shown for the certain objects to the specific user. On the other hand, the metadata of the ontology entity property B can be shown with no value shown, or alternatively no information regarding the ontology entity property B would be shown depending on the user role of the specific user.

4. Example Processes

(74) Each of FIGS. 7-9 is shown in simplified, schematic format for purposes of illustrating a clear example and other embodiments may include more, fewer, or different elements connected in various manners. Each of FIGS. 7-9 is intended to disclose an algorithm, plan, or outline that can be used to implement one or more computer programs or other software elements which when executed cause performing the functional improvements and technical advances that are described herein. Furthermore, the flow diagrams herein are described at the same level of detail that persons of ordinary skill in the art ordinarily use to communicate with one another about algorithms, plans, or specifications forming a basis of software programs that they plan to code or implement using their accumulated skill and knowledge.

(75) FIG. 7 illustrates an example process of indexing data changes to ontology-based databases in accordance with disclosed embodiments.

(76) In some embodiments, the server **102** is programmed or configured to represent a data source of one or more data sources by a dataset of the one or more datasets. The server **102** is programmed or configured to then add a record to the first mapping that associates the dataset with one or more ontology entity types of the plurality of ontology entity types, and create a transformation between a schema for the dataset and the one or more definitions for the one or more ontology entity types. In addition, the server **102** is programmed or configured to generate one or more ontology entities from the dataset using the transformation.

(77) In step **702**, the server **102** is programmed or configured to receive a set of data source updates for one or more datasets from the one or more data sources. In some embodiments, the server **102** is programmed or configured to receive original data from the data source, determine changes to the dataset from the original data, and add the changes to a changelog dataset that represents the set of data source updates.

(78) In step **704**, the server **102** is programmed or configured to receive one or more sets of user edits to an ontology. The ontology includes a definition for each ontology entity type of a plurality of ontology entity types and a plurality of ontology entities instantiated from the plurality of ontology entity types. At least one ontology entity of the plurality of ontology entities is represented in multiple forms respectively in multiple object databases of a plurality of object databases.

(79) In some embodiments, an ontology entity is an object or a link between two objects. In certain embodiments, a user edit of the one or more sets of user edits is a change to an ontology entity of the set of ontology entities. Each set of user edits of the one or more sets of user edits is to be applied as a single transaction to a user.

(80) In step **706**, the server **102** is programmed or configured to transform the set of data source updates to a list of updates to the ontology based on a first mapping between the one or more

datasets and the plurality of ontology entity types. In some embodiments, multiple datasets of the one or more datasets are mapped to a common ontology entity type in the first mapping.

(81) In step **708**, the server **102** is programmed or configured to merge the list of updates to the ontology with the one or more sets of user edits into a merged dataset. In some embodiments, the merging comprises joining the changelog dataset noted above and the one or more sets of user edits based on a primary key of a certain ontology entity. The merging further comprises storing the merged dataset in a memory, thereby materializing the one or more sets of user edits. In certain embodiments, the merging is performed when a size of the list of updates or a size of the one or more sets of user edits has exceeded a threshold or when a certain amount of time has passed.

(82) In step **710**, the server **102** is programmed or configured to generate index data from the merged dataset for one or more object databases of the plurality of object databases based on a second mapping between the plurality of ontology entity types and the plurality of object databases. In some embodiments, the plurality of object databases includes a graph database and a relational database respectively associated with different data representations and query operations. In certain embodiments, the second mapping associates each ontology entity type with at least one object database of the plurality of object databases.

(83) In some embodiments, the merged dataset includes changes to one or more ontology entities of a set of ontology entity types of the plurality of ontology entity types. The generating comprises, for a specific object database of the plurality of object databases, creating one or more index files from the changes to the one or more ontology entities for the set of ontology entity types that are associated with the specific object database, and the one or more index files are in a specific format required by the specific object database. In certain embodiments, each ontology entity type of the plurality of ontology entity types has one or more properties. The index data associates a value for a property of an ontology entity type of the plurality of ontology entity types with a list of ontology entities of the set of ontology entities.

(84) In step **712**, the server **102** is programmed or configured to transmit the index data to the one or more object databases.

(85) FIG. **8** illustrates a process of resolving conflicts in changes to ontology data in accordance with disclosed embodiments.

(86) In step **802**, the server **102** is programmed or configured to receive a set of data source updates for one or more datasets from one or more data sources.

(87) In step **804**, the server **102** is programmed or configured to receive one or more sets of user edits to an ontology. The ontology includes a definition for each ontology entity type of a plurality of ontology entity types and a plurality of ontology entities instantiated from the plurality of ontology entity types. In some embodiments, an ontology entity is an object or a link between two objects. In certain embodiments, at least one ontology entity of the plurality of ontology entities is represented in multiple forms respectively in multiple object databases of a plurality of object databases.

(88) In step **806**, In step **802**, the server **102** is programmed or configured to transform the set of data source updates to a list of updates to the ontology based on a first mapping between the one or more datasets and the plurality of ontology entity types.

(89) In step **808**, In step **802**, the server **102** is programmed or configured to merge the list of updates to the ontology with the one or more sets of user edits into a merged dataset. In some embodiments, the merging is performed when a size of the list of updates to the ontology or a size of the one or more sets of user edits has exceeded a threshold or when a certain amount of time has passed.

(90) In step **810**, In step **802**, the server **102** is programmed or configured to resolve conflicts in the merged dataset related to a first ontology entity type based on a first strategy that prioritizes a user edit over a data source update, comprising, after a first ontology entity of the first ontology entity type having a first primary key is modified with a first user edit, deleting the first ontology entity

from the merged dataset according to a first data source update.

(91) In some embodiments, the resolving further comprising after the deleting, recreating the first ontology entity having the first primary key according a second data source update in the merged dataset, and reapplying the first user edit to the first ontology entity. In certain embodiments, the server **102** is programmed to resolve conflicts in the merged dataset related to each ontology entity type that is a sub-type of the first ontology entity type based on the first strategy. In other embodiments, the server **102** is programmed to resolve conflicts in the merged dataset related to a second ontology entity type based on a second strategy that prioritizes a more recent change to the ontology over a less recent change.

(92) In some embodiments, the server **102** is programmed to resolve conflicts in the merged dataset related to a second ontology entity type by reviewing data in the merged dataset in a chronological order. The server **102** is further programmed to determine that a current user edit to a second ontology entity of the second ontology entity type in the merged dataset is to delete the second ontology entity, and delete the second ontology entity from the merged dataset based on the first strategy.

(93) In some embodiments, the server **102** is further programmed to determine that a current user edit to a third ontology entity of the second ontology entity type in the merged dataset is to create the third ontology entity and an earlier user edit to the third ontology entity was to delete the third ontology entity, and disregard any later data source update to the third ontology entity in the merged dataset.

(94) In some embodiments, the server **102** is further programmed to determine that a current user edit to a fourth ontology entity of the second ontology entity type in the merged dataset is to create the fourth ontology entity and no earlier user edit to the fourth ontology entity was to delete the fourth ontology entity, and disregard any later data source update to the fourth ontology entity in the merged dataset.

(95) In some embodiments, the server **102** is further programmed to determine that a current user edit to a fifth ontology entity of the second ontology entity type in the merged dataset is to modify a property of the fifth ontology entity, no earlier user edit to the fifth ontology entity was to create the fifth ontology entity, and the fifth ontology entity is represented in the one or more datasets, and disregard any later data source update to modify the property of the fifth ontology entity in the merged dataset.

(96) In some embodiments, the server **102** is further programmed to determine that a current user edit to a sixth ontology entity of the second ontology entity type in the merged dataset is to modify a property of the sixth ontology entity, no earlier user edit to the sixth ontology entity was to create the sixth ontology entity, and the sixth ontology entity is not represented in the one or more datasets, and reject the current user edit.

(97) FIG. 9 illustrates a process of managing access control for ontology data FIG. 9 illustrates a process of managing access control for ontology data in accordance with disclosed embodiments.

(98) In step **902**, the server **102** is programmed or configured to receive a set of data source updates for one or more datasets from one or more data sources.

(99) In step **904**, the server **102** is programmed or configured to transform the set of data source updates to a list of updates to an ontology based on a first mapping. The ontology includes a definition for each ontology entity type of a plurality of ontology entity types, the definition including one or more properties. The ontology also includes a plurality of ontology entities instantiated from the plurality of ontology entity types. The first mapping is between the one or more datasets and the plurality of ontology entity types. In some embodiments, an ontology entity is an object or a link between two objects.

(100) In some embodiments, the server **102** is programmed to represent a data source of the one or more data sources by a dataset of the one or more datasets. The server **102** is also programmed to add a record to the first mapping that associates the dataset with one or more ontology entity types

of the plurality of ontology entity types, and create a transformation between a schema for the dataset and one or more definitions for the one or more ontology entity types. In addition, the server **102** is programmed to generate one or more ontology entities from the dataset using the transformation.

(101) In step **906**, the server **102** is programmed or configured to manage a set of user accounts of a user account type having one or more attributes. In some embodiments, a specific attribute of the one or more attributes of the user account type corresponds to a user role of plurality of user roles. Each user role is associated with a distinct set of discovering, reading, writing, or administering permissions.

(102) In some embodiments, the server **102** is programmed to receive, for the dataset, a set of permissions controlling access to rows of the dataset by the set of user accounts. The server **102** is also programmed to create a specific security policy that applies to ontology entities of the one or more ontology entity types based on the set of permissions. In certain embodiments, the providing comprises applying the specific security policy before applying the first security policy.

(103) In some embodiments, the server **102** is programmed to create a first transformation mapping columns of a first dataset of the one or more datasets to a first set of properties of a certain ontology entity type of the plurality of ontology entity types, and create a second transformation mapping columns of a second dataset of the one or more datasets to a second set of properties of the certain ontology entity type. The server is programmed to then create a specific security policy that applies to the first set of properties of ontology entities of the certain ontology entity type based on a first set of permissions controlling access to rows of the first dataset by the set of user accounts, and create a particular security policy that applies to the second set of properties of the ontology entities of the certain ontology entity type based on a second set of permissions controlling access to rows of the second dataset by the set of user accounts.

(104) In step **908**, the server **102** is programmed or configured to cause presenting a first option to create a rule by choosing two items from an attribute of the user account type, a property of an ontology entity type, and a value, and a second option to combine a plurality of rules by choosing a logical operator from a plurality of logical operators. In some embodiments, the server **102** is programmed to cause presenting a third option to combine the two items by choosing a relational operator from a plurality of relational operators. In certain embodiments, the server **102** is programmed to cause presenting a fourth option to associate one or more permissions with the rule that are granted when the rule applies. In other embodiments, the one or more permissions allow no access to a first property of a specific ontology entity type of the plurality of ontology entity types and allowing access to a second property of the specific ontology entity type.

(105) In step **910**, the server **102** is programmed or configured to create a first security policy based on input data received via the first option and the second option. In some embodiments, the creating comprises determining a scope of each rule in the plurality of rules based on the associated one or more permissions, and determining a scope of the plurality of rules in combination based on the scope of each rule and the logical operator.

(106) In step **912**, the server **102** is programmed or configured to provide access to the ontology to user accounts according to the first security policy.

(107) In some embodiments, the list of updates to the ontology includes a list of changes to at least one ontology entity of the plurality of ontology entities. The server **102** is further programmed to, for an ontology entity of the at least one ontology entity, represent the list of updates that applies to the ontology entity in multiple forms to generate multiple representations respectively in multiple object databases. The providing then comprises applying the first security policy equally to the multiple representations.

5. Example Implementation

(108) According to one embodiment, the techniques described herein are implemented by at least one computing device. The techniques may be implemented in whole or in part using a

combination of at least one server computer and/or other computing devices that are coupled using a network, such as a packet data network. The computing devices may be hard-wired to perform the techniques, or may include digital electronic devices such as at least one application-specific integrated circuit (ASIC) or field programmable gate array (FPGA) that is persistently programmed to perform the techniques, or may include at least one general purpose hardware processor programmed to perform the techniques pursuant to program instructions in firmware, memory, other storage, or a combination. Such computing devices may also combine custom hard-wired logic, ASICs, or FPGAs with custom programming to accomplish the described techniques. The computing devices may be server computers, workstations, personal computers, portable computer systems, handheld devices, mobile computing devices, wearable devices, body mounted or implantable devices, smartphones, smart appliances, internetworking devices, autonomous or semi-autonomous devices such as robots or unmanned ground or aerial vehicles, any other electronic device that incorporates hard-wired and/or program logic to implement the described techniques, one or more virtual computing machines or instances in a data center, and/or a network of server computers and/or personal computers.

(109) FIG. **10** is a block diagram that illustrates an example computer system with which an embodiment may be implemented. In the example of FIG. **10**, a computer system **1000** and instructions for implementing the disclosed technologies in hardware, software, or a combination of hardware and software, are represented schematically, for example as boxes and circles, at the same level of detail that is commonly used by persons of ordinary skill in the art to which this disclosure pertains for communicating about computer architecture and computer systems implementations.

(110) Computer system **1000** includes an input/output (I/O) subsystem **1002** which may include a bus and/or other communication mechanism(s) for communicating information and/or instructions between the components of the computer system **1000** over electronic signal paths. The I/O subsystem **1002** may include an I/O controller, a memory controller and at least one I/O port. The electronic signal paths are represented schematically in the drawings, for example as lines, unidirectional arrows, or bidirectional arrows.

(111) At least one hardware processor **1004** is coupled to I/O subsystem **1002** for processing information and instructions. Hardware processor **1004** may include, for example, a general-purpose microprocessor or microcontroller and/or a special-purpose microprocessor such as an embedded system or a graphics processing unit (GPU) or a digital signal processor or Advanced RISC Machines (ARM) processor. Processor **1004** may comprise an integrated arithmetic logic unit (ALU) or may be coupled to a separate ALU.

(112) Computer system **1000** includes one or more units of memory **1006**, such as a main memory, which is coupled to I/O subsystem **1002** for electronically digitally storing data and instructions to be executed by processor **1004**. Memory **1006** may include volatile memory such as various forms of random-access memory (RAM) or other dynamic storage device. Memory **1006** also may be used for storing temporary variables or other intermediate information during execution of instructions to be executed by processor **1004**. Such instructions, when stored in non-transitory computer-readable storage media accessible to processor **1004**, can render computer system **1000** into a special-purpose machine that is customized to perform the operations specified in the instructions.

(113) Computer system **1000** further includes non-volatile memory such as read only memory (ROM) **1008** or other static storage device coupled to I/O subsystem **1002** for storing information and instructions for processor **1004**. The ROM **1008** may include various forms of programmable ROM (PROM) such as erasable PROM (EPROM) or electrically erasable PROM (EEPROM). A unit of persistent storage **1010** may include various forms of non-volatile RAM (NVRAM), such as flash memory, or solid-state storage, magnetic disk, or optical disk such as CD-ROM or DVD-ROM, and may be coupled to I/O subsystem **1002** for storing information and instructions. Storage **1010** is an example of a non-transitory computer-readable medium that may be used to store

instructions and data which when executed by the processor **1004** cause performing computer-implemented methods to execute the techniques herein.

(114) The instructions in memory **1006**, ROM **1008** or storage **1010** may comprise one or more sets of instructions that are organized as modules, methods, objects, functions, routines, or calls. The instructions may be organized as one or more computer programs, operating system services, or application programs including mobile apps. The instructions may comprise an operating system and/or system software; one or more libraries to support multimedia, programming or other functions; data protocol instructions or stacks to implement Transmission Control Protocol/Internet Protocol (TCP/IP), Hypertext Transfer Protocol (HTTP) or other communication protocols; file processing instructions to interpret and render files coded using HTML, XML, Joint Photographic Experts Group (JPEG), Moving Picture Experts Group (MPEG) or Portable Network Graphics (PNG); user interface instructions to render or interpret commands for a GUI, command-line interface or text user interface; application software such as an office suite, internet access applications, design and manufacturing applications, graphics applications, audio applications, software engineering applications, educational applications, games or miscellaneous applications. The instructions may implement a web server, web application server or web client. The instructions may be organized as a presentation layer, application layer and data storage layer such as a relational database system using structured query language (SQL) or NoSQL, an object store, a graph database, a flat file system or other data storage.

(115) Computer system **1000** may be coupled via I/O subsystem **1002** to at least one output device **1012**. In one embodiment, output device **1012** is a digital computer display. Examples of a display that may be used in various embodiments include a touch screen display or a light-emitting diode (LED) display or a liquid crystal display (LCD) or an e-paper display. Computer system **1000** may include other type(s) of output devices **1012**, alternatively or in addition to a display device. Examples of other output devices **1012** include printers, ticket printers, plotters, projectors, sound cards or video cards, speakers, buzzers or piezoelectric devices or other audible devices, lamps or LED or LCD indicators, haptic devices, actuators, or servos.

(116) At least one input device **1014** is coupled to I/O subsystem **1002** for communicating signals, data, command selections or gestures to processor **1004**. Examples of input devices **1014** include touch screens, microphones, still and video digital cameras, alphanumeric and other keys, keypads, keyboards, graphics tablets, image scanners, joysticks, clocks, switches, buttons, dials, slides, and/or various types of sensors such as force sensors, motion sensors, heat sensors, accelerometers, gyroscopes, and inertial measurement unit (IMU) sensors and/or various types of transceivers such as wireless, such as cellular or Wi-Fi, radio frequency (RF) or infrared (IR) transceivers and Global Positioning System (GPS) transceivers.

(117) Another type of input device is a control device **1016**, which may perform cursor control or other automated control functions such as navigation in a graphical interface on a display screen, alternatively or in addition to input functions. Control device **1016** may be a touchpad, a mouse, a trackball, or cursor direction keys for communicating direction information and command selections to processor **1004** and for controlling cursor movement on the output device **1012**. The input device may have at least two degrees of freedom in two axes, a first axis (e.g., x) and a second axis (e.g., y), that allows the device to specify positions in a plane. Another type of input device is a wired, wireless, or optical control device such as a joystick, wand, console, steering wheel, pedal, gearshift mechanism or other type of control device. An input device **1014** may include a combination of multiple different input devices, such as a video camera and a depth sensor.

(118) In another embodiment, computer system **1000** may comprise an internet of things (IoT) device in which one or more of the output device **1012**, input device **1014**, and control device **1016** are omitted. Or, in such an embodiment, the input device **1014** may comprise one or more cameras, motion detectors, thermometers, microphones, seismic detectors, other sensors or detectors,

measurement devices or encoders and the output device **1012** may comprise a special-purpose display such as a single-line LED or LCD display, one or more indicators, a display panel, a meter, a valve, a solenoid, an actuator or a servo.

(119) When computer system **1000** is a mobile computing device, input device **1014** may comprise a global positioning system (GPS) receiver coupled to a GPS module that is capable of triangulating to a plurality of GPS satellites, determining and generating geo-location or position data such as latitude-longitude values for a geophysical location of the computer system **1000**.

Output device **1012** may include hardware, software, firmware, and interfaces for generating position reporting packets, notifications, pulse or heartbeat signals, or other recurring data transmissions that specify a position of the computer system **1000**, alone or in combination with other application-specific data, directed toward host computer **1024** or server **1030**.

(120) Computer system **1000** may implement the techniques described herein using customized hard-wired logic, at least one ASIC or FPGA, firmware and/or program instructions or logic which when loaded and used or executed in combination with the computer system causes or programs the computer system to operate as a special-purpose machine. According to one embodiment, the techniques herein are performed by computer system **1000** in response to processor **1004** executing at least one sequence of at least one instruction contained in main memory **1006**. Such instructions may be read into main memory **1006** from another storage medium, such as storage **1010**.

Execution of the sequences of instructions contained in main memory **1006** causes processor **1004** to perform the process steps described herein. In alternative embodiments, hard-wired circuitry may be used in place of or in combination with software instructions.

(121) The term “storage media” as used herein refers to any non-transitory media that store data and/or instructions that cause a machine to operate in a specific fashion. Such storage media may comprise non-volatile media and/or volatile media. Non-volatile media includes, for example, optical or magnetic disks, such as storage **1010**. Volatile media includes dynamic memory, such as memory **1006**. Common forms of storage media include, for example, a hard disk, solid state drive, flash drive, magnetic data storage medium, any optical or physical data storage medium, memory chip, or the like.

(122) Storage media is distinct from but may be used in conjunction with transmission media. Transmission media participates in transferring information between storage media. For example, transmission media includes coaxial cables, copper wire and fiber optics, including the wires that comprise a bus of I/O subsystem **1002**. Transmission media can also take the form of acoustic or light waves, such as those generated during radio-wave and infra-red data communications.

(123) Various forms of media may be involved in carrying at least one sequence of at least one instruction to processor **1004** for execution. For example, the instructions may initially be carried on a magnetic disk or solid-state drive of a remote computer. The remote computer can load the instructions into its dynamic memory and send the instructions over a communication link such as a fiber optic or coaxial cable or telephone line using a modem. A modem or router local to computer system **1000** can receive the data on the communication link and convert the data to be read by computer system **1000**. For instance, a receiver such as a radio frequency antenna or an infrared detector can receive the data carried in a wireless or optical signal and appropriate circuitry can provide the data to I/O subsystem **1002** such as place the data on a bus. I/O subsystem **1002** carries the data to memory **1006**, from which processor **1004** retrieves and executes the instructions. The instructions received by memory **1006** may optionally be stored on storage **1010** either before or after execution by processor **1004**.

(124) Computer system **1000** also includes a communication interface **1018** coupled to I/O subsystem **1002**. Communication interface **1018** provides a two-way data communication coupling to network link(s) **1020** that are directly or indirectly connected to at least one communication network, such as a network **1022** or a public or private cloud on the Internet. For example, communication interface **1018** may be an Ethernet networking interface, integrated-services digital

network (ISDN) card, cable modem, satellite modem, or a modem to provide a data communication connection to a corresponding type of communications line, for example an Ethernet cable or a metal cable of any kind or a fiber-optic line or a telephone line. Network **1022** broadly represents a LAN, WAN, campus network, internetwork, or any combination thereof. Communication interface **1018** may comprise a LAN card to provide a data communication connection to a compatible LAN, or a cellular radiotelephone interface that is wired to send or receive cellular data according to cellular radiotelephone wireless networking standards, or a satellite radio interface that is wired to send or receive digital data according to satellite wireless networking standards. In any such implementation, communication interface **1018** sends and receives electrical, electromagnetic, or optical signals over signal paths that carry digital data streams representing various types of information.

(125) Network link **1020** typically provides electrical, electromagnetic, or optical data communication directly or through at least one network to other data devices, using, for example, satellite, cellular, Wi-Fi, or BLUETOOTH technology. For example, network link **1020** may provide a connection through a network **1022** to a host computer **1024**.

(126) Furthermore, network link **1020** may provide a connection through network **1022** or to other computing devices via internetworking devices and/or computers that are operated by an Internet Service Provider (ISP) **1026**. ISP **1026** provides data communication services through a world-wide packet data communication network represented as internet **1028**. A server **1030** may be coupled to internet **1028**. Server **1030** broadly represents any computer, data center, virtual machine, or virtual computing instance with or without a hypervisor, or computer executing a containerized program system such as DOCKER or KUBERNETES. Server **1030** may represent an electronic digital service that is implemented using more than one computer or instance and that is accessed and used by transmitting web services requests, Uniform Resource Locator (URL) strings with parameters in HTTP payloads, API calls, app services calls, or other service calls. Computer system **1000** and server **1030** may form elements of a distributed computing system that includes other computers, a processing cluster, server farm or other organization of computers that cooperate to perform tasks or execute applications or services. Server **1030** may comprise one or more sets of instructions that are organized as modules, methods, objects, functions, routines, or calls. The instructions may be organized as one or more computer programs, operating system services, or application programs including mobile apps. The instructions may comprise an operating system and/or system software; one or more libraries to support multimedia, programming or other functions; data protocol instructions or stacks to implement TCP/IP, HTTP or other communication protocols; file format processing instructions to interpret or render files coded using HTML, XML, JPEG, MPEG or PNG; user interface instructions to render or interpret commands for a GUI, command-line interface or text user interface; application software such as an office suite, internet access applications, design and manufacturing applications, graphics applications, audio applications, software engineering applications, educational applications, games or miscellaneous applications. Server **1030** may comprise a web application server that hosts a presentation layer, application layer and data storage layer such as a relational database system using SQL or NoSQL, an object store, a graph database, a flat file system or other data storage.

(127) Computer system **1000** can send messages and receive data and instructions, including program code, through the network(s), network link **1020** and communication interface **1018**. In the Internet example, a server **1030** might transmit a requested code for an application program through Internet **1028**, ISP **1026**, local network **1022** and communication interface **1018**. The received code may be executed by processor **1004** as it is received, and/or stored in storage **1010**, or other non-volatile storage for later execution.

(128) The execution of instructions as described in this section may implement a process in the form of an instance of a computer program that is being executed, and consisting of program code and its current activity. Depending on the operating system (OS), a process may be made up of

multiple threads of execution that execute instructions concurrently. In this context, a computer program is a passive collection of instructions, while a process may be the actual execution of those instructions. Several processes may be associated with the same program; for example, opening up several instances of the same program often means more than one process is being executed. Multitasking may be implemented to allow multiple processes to share processor **1004**. While each processor **1004** or core of the processor executes a single task at a time, computer system **1000** may be programmed to implement multitasking to allow each processor to switch between tasks that are being executed without having to wait for each task to finish. In an embodiment, switches may be performed when tasks perform input/output operations, when a task indicates that it can be switched, or on hardware interrupts. Time-sharing may be implemented to allow fast response for interactive user applications by rapidly performing context switches to provide the appearance of concurrent execution of multiple processes simultaneously. In an embodiment, for security and reliability, an operating system may prevent direct communication between independent processes, providing strictly mediated and controlled inter-process communication functionality.

6. Extensions and Alternatives

(129) In the foregoing specification, embodiments of the disclosure have been described with reference to numerous specific details that may vary from implementation to implementation. The specification and drawings are, accordingly, to be regarded in an illustrative rather than a restrictive sense. The sole and exclusive indicator of the scope of the disclosure, and what is intended by the applicants to be the scope of the disclosure, is the literal and equivalent scope of the set of claims that issue from this application, in the specific form in which such claims issue, including any subsequent correction.

Claims

1. A computer-implemented method of resolving conflicts in changes to ontology data, comprising: receiving a set of data source updates for one or more datasets from one or more data sources; receiving one or more sets of user edits to an ontology, the ontology including a definition for each ontology entity type of a plurality of ontology entity types, the ontology including a plurality of ontology entities instantiated from the plurality of ontology entity types; transforming the set of data source updates to a list of updates to the ontology based on a first mapping between the one or more datasets and the plurality of ontology entity types; merging the list of updates to the ontology with the one or more sets of user edits into a merged dataset; resolving conflicts in the merged dataset related to a first ontology entity type based on a first strategy that prioritizes a user edit over a data source update, comprising, after a first ontology entity of the first ontology entity type having a first primary key is modified with a first user edit, deleting the first ontology entity from the merged dataset according to a first data source update, wherein the method is implemented using one or more processors.
2. The computer-implemented method of claim 1, the resolving further comprising: after the deleting, recreating the first ontology entity having the first primary key according a second data source update in the merged dataset; reapplying the first user edit to the first ontology entity.
3. The computer-implemented method of claim 1, further comprising resolving conflicts in the merged dataset related to a second ontology entity type by reviewing data in the merged dataset in a chronological order, comprising: determining that a current user edit to a second ontology entity of the second ontology entity type in the merged dataset is to delete the second ontology entity; deleting the second ontology entity from the merged dataset based on the first strategy.
4. The computer-implemented method of claim 1, further comprising resolving conflicts in the merged dataset related to a second ontology entity type by reviewing data in the merged dataset in a chronological order, comprising: determining that a current user edit to a second ontology entity of the second ontology entity type in the merged dataset is to create the second ontology entity and an

earlier user edit to the second ontology entity was to delete the second ontology entity; disregarding any later data source update to the second ontology entity in the merged dataset.

5. The computer-implemented method of claim 1, further comprising resolving conflicts in the merged dataset related to a second ontology entity type by reviewing data in the merged dataset in a chronological order, comprising: determining that a current user edit to a second ontology entity of the second ontology entity type in the merged dataset is to create the second ontology entity and no earlier user edit to the second ontology entity was to delete the second ontology entity; disregarding any later data source update to the second ontology entity in the merged dataset.

6. The computer-implemented method of claim 1, further comprising resolving conflicts in the merged dataset related to a second ontology entity type by reviewing data in the merged dataset in a chronological order, comprising: determining that a current user edit to a second ontology entity of the second ontology entity type in the merged dataset is to modify a property of the second ontology entity, no earlier user edit to the second ontology entity was to create the second ontology entity, and the second ontology entity is represented in the one or more datasets; disregarding any later data source update to modify the property of the second ontology entity in the merged dataset.

7. The computer-implemented method of claim 1, further comprising resolving conflicts in the merged dataset related to a second ontology entity type by reviewing data in the merged dataset in a chronological order, comprising: determining that a current user edit to a second ontology entity of the second ontology entity type in the merged dataset is to modify a property of the second ontology entity, no earlier user edit to the second ontology entity was to create the second ontology entity, and the second ontology entity is not represented in the one or more datasets; rejecting the current user edit.

8. The computer-implemented method of claim 1, further comprising resolving conflicts in the merged dataset related to each ontology entity type that is a sub-type of the first ontology entity type based on the first strategy.

9. The computer-implemented method of claim 1, further comprising resolving conflicts in the merged dataset related to a second ontology entity type based on a second strategy that prioritizes a more recent change to the ontology over a less recent change.

10. The computer-implemented method of claim 1, the merging being performed when a size of the list of updates to the ontology or a size of the one or more sets of user edits has exceeded a threshold or when a certain amount of time has passed.

11. The computer-implemented method of claim 1, an ontology entity being an object or a link between two objects.

12. The computer-implemented method of claim 1, at least one ontology entity of the plurality of ontology entities being represented in multiple forms respectively in multiple object databases of a plurality of object databases.

13. The computer-implemented method of claim 1, further comprising resolving conflicts in the merged dataset related to a second ontology entity type based on a second strategy that prioritizes a more recent change to the ontology over a less recent change.

14. A system for resolving conflicts in changes to ontology data, comprising: a memory; one or more processors coupled to the memory and configured to perform: receiving a set of data source updates for one or more datasets from one or more data sources; receiving one or more sets of user edits to an ontology, the ontology including a definition for each ontology entity type of a plurality of ontology entity types, the ontology including a plurality of ontology entities instantiated from the plurality of ontology entity types; transforming the set of data source updates to a list of updates to the ontology based on a first mapping between the one or more datasets and the plurality of ontology entity types; merging the list of updates to the ontology with the one or more sets of user edits into a merged dataset; resolving conflicts in the merged dataset related to a first ontology entity type based on a first strategy that prioritizes a user edit over a data source update, comprising, after a first ontology entity of the first ontology entity type having a first primary key

is modified with a first user edit, deleting the first ontology entity from the merged dataset according to a first data source update.

15. The system of claim 14, the resolving further comprising: after the deleting, recreating the first ontology entity having the first primary key according a second data source update in the merged dataset; reapplying the first user edit to the first ontology entity.

16. The system of claim 14, the one or more processors further configured to perform resolving conflicts in the merged dataset related to a second ontology entity type by reviewing data in the merged dataset in a chronological order, comprising: determining that a current user edit to a second ontology entity of the second ontology entity type in the merged dataset is to delete the second ontology entity; deleting the second ontology entity from the merged dataset based on the first strategy.

17. The system of claim 14, the one or more processors further configured to perform resolving conflicts in the merged dataset related to a second ontology entity type by reviewing data in the merged dataset in a chronological order, comprising: determining that a current user edit to a second ontology entity of the second ontology entity type in the merged dataset is to create the second ontology entity and an earlier user edit to the second ontology entity was to delete the second ontology entity; disregarding any later data source update to the second ontology entity in the merged dataset.

18. The system of claim 14, the one or more processors further configured to perform resolving conflicts in the merged dataset related to a second ontology entity type by reviewing data in the merged dataset in a chronological order, comprising: determining that a current user edit to a second ontology entity of the second ontology entity type in the merged dataset is to modify a property of the second ontology entity, no earlier user edit to the second ontology entity was to create the second ontology entity, and the second ontology entity is represented in the one or more datasets; disregarding any later data source update to modify the property of the second ontology entity in the merged dataset.

19. The system of claim 14, the one or more processors further configured to perform resolving conflicts in the merged dataset related to a second ontology entity type by reviewing data in the merged dataset in a chronological order, comprising: determining that a current user edit to a second ontology entity of the second ontology entity type in the merged dataset is to modify a property of the second ontology entity, no earlier user edit to the second ontology entity was to create the second ontology entity, and the second ontology entity is not represented in the one or more datasets; rejecting the current user edit.

20. A non-transitory, computer-readable storage medium storing one or more sequences of instructions which when executed cause one or more processors to perform: receiving a set of data source updates for one or more datasets from one or more data sources; receiving one or more sets of user edits to an ontology, the ontology including a definition for each ontology entity type of a plurality of ontology entity types, the ontology including a plurality of ontology entities instantiated from the plurality of ontology entity types; transforming the set of data source updates to a list of updates to the ontology based on a first mapping between the one or more datasets and the plurality of ontology entity types; merging the list of updates to the ontology with the one or more sets of user edits into a merged dataset; resolving conflicts in the merged dataset related to a first ontology entity type based on a first strategy that prioritizes a user edit over a data source update, comprising, after a first ontology entity of the first ontology entity type having a first primary key is modified with a first user edit, deleting the first ontology entity from the merged dataset according to a first data source update.
